

WASP-36b

R-1163

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_210114 · created 2026-07-31T02:37:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459228.7664 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.09 +/- 0.047
Transit depth (Rp/Rs)^2	0.8 +/- 0.85 [%]
Orbital Inclination (inc)	79.66 +/- 2.89
Airmass coefficient 1 (a1)	0.96 +/- 0.47
Airmass coefficient 2 (a2)	0.01 +/- 0.2
Scatter in the residuals of the lightcurve fit is	48.16 %
Best Comparison Star	#3 - [544, 37]
Optimal Aperture	2.17
Optimal Annulus	7.5
Transit Duration (day)	0.029 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.09 $\pm$ 0.047 vs 0.1368 (deviation 0.98 σ)
Duration fit vs published	0.029 d vs 0.0758 d (deviation 1.54 σ)
Mid-time O-C	-2.3 min
Depth SNR	1.9
Residual scatter	48.16 %

Light curve

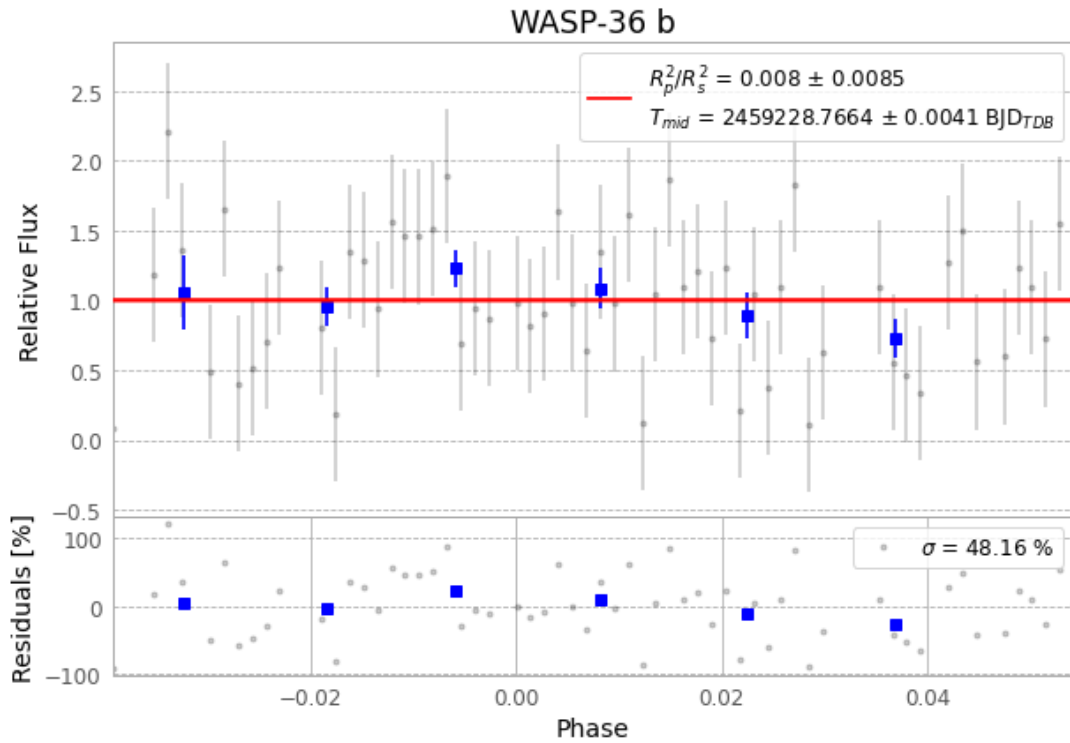


Figure 1 — FinalLightCurve_WASP-36 b_2021-01-13.png (SHA-256 ec768c1b8d033788...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [493, 388], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e1e770d7357da0e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), WASP-36210114044812.FITS → WASP-36210114081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-210114.log (160d94e965ee...), FinalLightCurve_WASP-36 b_2021-01-13.png (ec768c1b8d03...), FinalLightCurve_WASP-36 b_2021-01-13.csv (7328638ea867...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:37Z.